



**Faculty of Science and Arts
Department of Chemistry
Biochemistry Lab (CHEM 266)**

Experiment 4

GENERAL PROPERTIES OF PROTEINS

Pre-Lab Questions:

1. A protein formed from 15 amino acids, what is the number of peptide bonds in that protein?
2. Is Biuret test is specific for only proteins?
3. In most cases, what is the general charge of proteins in acidic media? Basic media?
4. Define the heavy metals? Why heavy metals are highly toxic to our body?
5. Give examples on proteins in our body and indicate their role?

Precautions:

1. Concentrated nitric acid will be used in this experiment; you have to deal with this strong acid **carefully** because it causes serious burns if you spill it on your skin.

Objectives:

1. Identify the peptide bond in proteins and know how that bond is formed
2. Investigate some physical and chemical properties of proteins.
3. Acquaint you with different protein color and precipitation reactions.
4. Explain the effect of each denaturing agent on proteins.

Introduction:

Proteins are high molecular weight, nitrogenous compounds that are formed from C, H, O, and N. proteins composed of amino acids, which are linked together by a linkage called **peptide linkage**. In peptide linkage, the carboxylic group from one amino acid is linked via a covalent bond with the amine group of the second amino acid and so on (**Figure 1**).

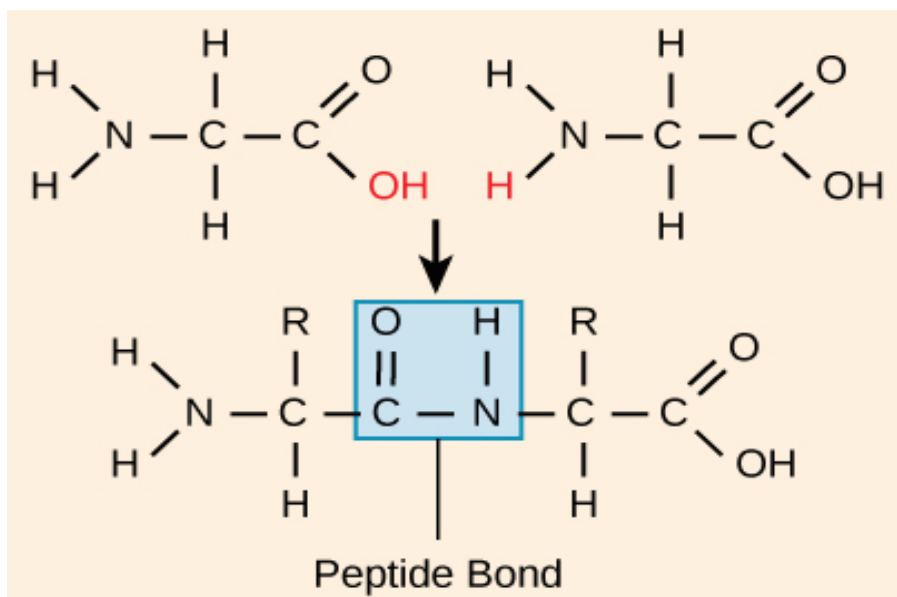


Figure 1. Peptide bond is formed between two amino acids through a dehydration reaction, first amino acid loses the hydroxyl group from the carboxyl group and the second amino acid loses hydrogen from the amino group, finally nitrogen substitutes in a place of the hydroxyl group leading to the formation of the peptide bond.

<https://biologydictionary.net/peptide-bond/> [Accessed on 2/13/2018]

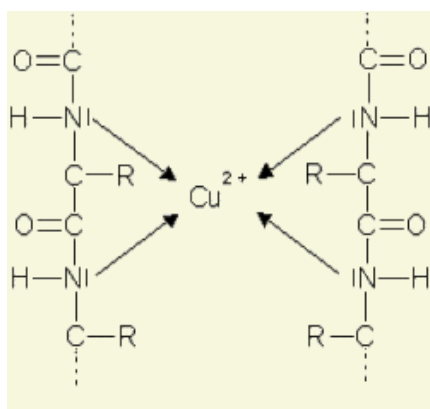
As mentioned previously, proteins play an important role in many biochemical and physiological processes inside the body, such as the formation of hemoglobin, antibodies, buffering system of the blood and many others. Therefore, performing some general reactions to test proteins and understanding the factors that affect on the structure and the function of them is critical.

Many agents can change the structure, as a result, affect on the function of proteins. Those factors are called denaturing agents and they can be classified based on their effect on proteins to reversible or irreversible denaturizing factors, in some cases we observe the denaturation in the form of precipitation of the proteins. The **denaturing agents** might be:

- **Physical agents:** high temperature, high-pressure, and X-ray.
- **Chemical agents:** Strong acids, strong alkali, organic solvents, and heavy metals.

A. General tests for proteins: The Biuret test:

In Biuret assay proteins; specifically peptide bond reacts with copper sulfate (CuSO_4) in an alkaline medium (that why a strong base such as NaOH) to form a violet complex. At least two peptide bonds are required to form the complex. The intensity of the violet color is directly proportional to the number of peptide bonds; this means that as the number of peptide bonds increases, the intensity of the blue-violet color increases (**Figure2**).



Biuret complex

Figure 2. Biuret complex Copper will form a coordination bond with the lone pair of electron of nitrogen; this leads to the formation of violet Biuret complex.

<https://www.coursehero.com/file/19996957/chemistry-of-proteins/> [Accessed on 2/13/2018]

Experimental procedure:

Materials needed

1. Copper sulfate (10 g/L $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ solution)
2. Protein solutions (5g/L of Casein dissolved in little diluted NaOH , 5 g/L of Albumin, and Gelatin protein dissolved in saline.
3. Sodium hydroxide (10M NaOH)

Procedure

1. In a test tube add 5 drops of copper sulfate solution to 2 mL of the protein solution followed by 2 mL sodium hydroxide solution.
2. Mix thoroughly and observe the color produced.

B. Denaturation reactions of proteins:

1. Denaturation by heat

Heat or high temperature is one of the factors that can disrupt and possibly destruct the protein structure. This process causes an irreversible effect on proteins. Proteins might be affected because of the effect of the heat on the hydrogen bond, salt bridge, disulfide bond, and others in proteins.

Experimental procedure:

Materials needed

1. The following proteins: 5g/L of Casein dissolved in little diluted NaOH, 5 g/L of Albumin, and Gelatin protein dissolved in saline.
2. 1M NaOH solution.
3. 1M hydrochloric acid (HCl).
4. Concentrated Nitric acid (HNO₃).
5. Boiling water bath.

Procedure

1. Take 2 ml of one protein in three test tubes and add 0.5 mL HCl, 0.5 mL NaOH, and 0.5 mL of water to these tubes.
2. Place the tubes in a boiling water bath for 10 minutes.
3. Remove the tubes from the boiling water bath then cool them to room temperature.

2. Denaturation by extreme pH

Similar to heat, extreme pH can disrupt and possibly destruct the protein structure. For example, in basic medium protein loose protons, mainly from $-\text{NH}_3^+$. In acidic medium protein accept proton $-\text{COO}^-$ is the accepting group. The change in the charge of protein leads to its destruction and consequently, affects the function of the protein.

Experimental procedure:

Materials needed:

1. The following proteins: 5g/L of Casein dissolved in little diluted NaOH, 5 g/L of Albumin, and Gelatin protein dissolved in saline.
2. Concentrated Nitric acid (HNO₃).

Procedure:

1. In a test tube pour 2 mL protein solution.
2. By angle of 45° add 2 mL of concentrated nitric acid to the inner side of the tube.
3. Upside the tube and realize the formation of a yellow precipitate as a junction between the protein layer and the acid layer.

3. Precipitation by heavy metal

At pH 7 or slightly basic medium, proteins usually have a negative charge, adding heavy metals such as Hg^{+2} , Pb^{+2} , or Cu^{+2} , counteracts the negative charge of protein, which lead to neutralizing them and precipitation happen.

Adding an excess of the positively charged heavy metal leads to solvation. Solvation is process by which the positive ion is surrounded by the solvent.

Experimental procedure:

Materials needed:

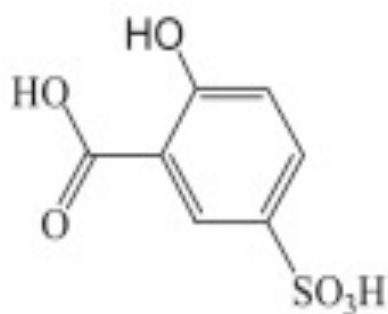
1. Proteins as before.
2. 0.1 M solutions of copper sulfate (CuSO_4), lead acetate ($\text{C}_4\text{H}_6\text{O}_4\text{Pb}$), mercuric nitrate ($\text{Hg}(\text{NO}_3)_2$) to give heavy metal ions.

Procedure:

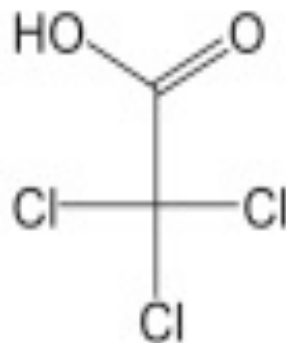
1. Add few drops of heavy metal solutions to 2 mL of the protein solution and realize the precipitation.
2. Add excess reagent and realize the dissolution of the precipitate. Why?

4. Precipitation by acidic reagent

As mentioned previously, acidic reagents destruct the proteins, which leads to precipitation. Sulfosalicylic acid and Trichloroacetic acid are two examples on the acids that have a negative charge (**figure 3**). At low pH proteins usually have a positive charge, therefore acids neutralize the protein charge, which leads to the formation of the insoluble precipitation.



5-sulfosalicylic acid



trichloroacetic acid

Figure 3. 5-sulfosalicylic acid and trichloroacetic acid chemical structure.

Experimental procedure:

Materials needed:

1. 5 g/L Albumin solution
2. Diluted NaOH solution
3. Acidic reagents; 20% [w/v] sulfosalicylic acid and 20% [W/V] trichloroacetic acid.

Procedure:

1. Add 5 drop of acidic reagent to 1 or 2 mL of the protein solution (albumin) and realize the precipitate.
2. Add an excess of acidic reagent. What is the effect of adding an excess of the acidic reagent?

Data analysis:

1. Write the report of this experiment in your own words.
2. Try to write simple formula for each reaction you did.
3. In the purpose section, write a purpose and conclusion for the part you did, in another word the purpose should match the conclusion.

References:

1. Introductory organic and biochemistry laboratory manual, Kansas State University, revised 2009.
2. <http://www1.mans.edu.eg/FacMed/english/dept/biochemistry/pdf/PROTEIN.pdf>

3. <https://www.coursehero.com/file/19996957/chemistry-of-proteins/>
4. Jin, S., Zhao, Y., Liu, B., Jin, X., Zhang, H., Wen, X., ... & Wang, D. (2015). Hydrogen bonded supramolecular structures of eight organic salts based on 2, 6-diaminopyridine, and organic acids. *Journal of Molecular Structure*, 1099, 601-615.



**Faculty of Science and Arts
Applied Chemistry
Biochemistry Lab (CHEM 266)
Experiment 4**

General properties of Proteins

Report Author:

Lab Partner:

Date Due:

Date Turned In:

Unknown #:

Unknown Identification:

Purpose:

Data and calculations:

A: Biuret test,

Protein	Albumin	Gelatin	Casein
Color			

B: denaturation of proteins:

1) Denaturation of proteins by temperature

Media	H ₂ O	HCL	NaOH
Observation			

2) Effect of addition of concentrated nitric acid:

C: Denaturation by heavy metal ions:

Effect of addition of Cu²⁺ to the protein solution:

Effect of addition of excess Cu²⁺ to the protein solution:

D: Denaturation by acidic reagent:

Effect of addition of acidic reagent to the protein solution:

Effect of addition of excess acidic reagent to the protein solution:

Conclusion:

Errors and problems you encountered and their possible remedies: